

# Module 2 Programming Part II



## In-class Challenge Description:

In this exercise challenge, you will write a Java GUI program that allows the user to enter personal information, including name, address, city, state, and zip code. The program should provide two main functions: saving the entered information to a text file and reading the data from the text file. This challenge tests your ability to create a functional GUI application and perform file I/O operations.

## Challenge Requirements:

Your Java GUI program should meet the following requirements:

### 1. User Interface:

- Create a graphical user interface (GUI) using Java Swing or JavaFX.
- Include text fields and labels for the following information:
  - Name
  - Address
  - City
  - State
  - Zip code
- Add two buttons:
  - A "Save" button to save the entered data to a text file.
  - A "Load" button to read data from the text file and display it in the text fields.

### 2. File I/O:

- Implement functionality to save the user-entered information to a text file. Each time the "Save" button is clicked, the information should be appended to the file.

- Implement functionality to read and display the data from the text file when the "Load" button is clicked. Display the information in the respective text fields.

### 3. Validation:

- Implement basic validation to ensure that all fields are filled in before saving the data.

### 4. User Feedback:

- Provide appropriate feedback to the user:
  - Inform the user when data is successfully saved.
  - Inform the user when data is successfully loaded.
  - Handle potential errors, such as file not found or file I/O errors, gracefully, and provide informative error messages to the user.

### 5. User-Friendly Interface:

- Ensure the GUI is user-friendly and intuitive, with clear labels and error messages.

## Challenge Grading Rubric:

The challenge will be graded based on the following criteria, with a maximum score of 100 points:

- **User Interface (25 points):**

- The GUI is well-organized, with clear labels for each input field.
- The "Save" and "Load" buttons are labeled appropriately.
- The GUI is visually appealing and user-friendly.

- **File I/O and Functionality (25 points):**

- The program can save user data to a text file, with each entry separated or organized appropriately.
- The program can read data from the text file and populate the respective text fields.
- The program handles file I/O errors and provides informative error messages to the user.

- **Validation (20 points):**

- The program validates that all fields are filled before saving.

- Invalid or incomplete entries do not lead to file I/O errors.
- **User Feedback (15 points):**
  - The program provides clear and informative feedback to the user when data is saved or loaded.
  - The program handles potential errors gracefully and communicates them to the user.
- **Code Quality and Structure (15 points):**
  - The code is well-organized and follows best practices in terms of structure, naming conventions, and comments.
  - The code is free from unnecessary redundancies and follows Java coding standards.
- **Extra Credit (up to 10 points):**
  - Additional features or enhancements beyond the basic requirements that demonstrate creativity and ingenuity.

## Submission Guidelines:

- Submit your Java source code file and a screenshot.
- Ensure that your code is well-commented to explain the functionality and usage of different parts of the program.

Good luck with the challenge!

Points 100

Submitting a file upload

| Due    | For      | Available from | Until             |
|--------|----------|----------------|-------------------|
| Sep 21 | Everyone | -              | Oct 22 at 11:59pm |